

Performance Comparison of Top Groq API Models

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Models:

- GPT-OSS-120B
- Llama 3.3 70B
- Qwen 3 32B

code

models.py > ...

```
1  from groq import Groq
2  import pandas as pd
3  import os
4  import time
5  import matplotlib.pyplot as plt
6
7  client = Groq(api_key=os.getenv("groq_api_key"))
8  conversation = [
9      {
10         "role": "system",
11         "content": "You are a good data analyst. Explain datasets in simple language and provide business insights."
12     }
13 ]
14
15 MODELS = {
16     "GPT-OSS-120B": "openai/gpt-oss-120b",
17     "Llama-3.3-70B": "llama-3.3-70b-versatile",
18     "Qwen-3-32B": "qwen/qwen3-32b"
19 }
20
21
22 def ask_model(model_name, messages):
23     start=time.time()
24
25     response = client.chat.completions.create(
26         model=model_name,
27         messages=messages,
28         max_completion_tokens=1000,
29         temperature=0.7
30     )
31     end=time.time()
32     time_taken=round(end-start,2)
33     return response.choices[0].message.content,time_taken
34
35
```

Welcome.py > ...

```
34
35
36 def compare_models(prompt):
37     """
38     Compare all models on the same prompt.
39     """
40
41     # Add the user's latest message
42     conversation.append(
43         {
44             "role": "user",
45             "content": prompt
46         }
47     )
48
49     results = {}
50
51     for name, model in MODELS.items():
52         answer, time_t = ask_model(model, conversation)
53         results[name] = {"answer": answer, "time": time_t}
54     conversation.append(
55         {
56             "role": "assistant",
57             "content": results["GPT-05S-120B"],
58         }
59     )
60
61     return results
```

models.py > ...

```
62
63 if __name__ == "__main__":
64
65     df = pd.read_csv(
66         r"C:\Users\nani3\Desktop\❤️\csv\sleep_cycle_productivity.csv"
67     )
68
69     summary = df.describe(include="all").to_string()
70
71     prompt = f"""
72 Here are statistics from a dataset.
73 {summary}
74 Explain the data in a simple way.
75 Identify trends,
76 Identify anomalies,
77 Give business insights,
78 and give me the total information in only 10 lines
79 """
80     responses = compare_models(prompt)
81
82
83     for model, result in responses.items():
84         print("=" * 100)
85         print(model)
86         print("=" * 100)
87         print(f"Time Taken : {result['time']} seconds\n")
88         print(result["answer"])
89     names=list(responses.keys())
90     time=[i['time'] for i in responses.values()]
91     plt.bar(names,time)
92     plt.xlabel('models')
93     plt.ylabel('time')
94     plt.title('models time')
95     plt.show()
```

summary from sleep_cycle_productivity.csv
(5000 rows of data)

=====

Qwen-3-32B

=====

Time Taken : 2.79 seconds

<think>

Okay, let's start by looking at the dataset provided. The user wants a simple explanation, trends, anomalies, business insights, and a summary in 10 lines. First, the dataset has 5000 entries with details on sleep, exercise, caffeine, screen time, work hours, productivity, mood, and stress. The unique date count is 366, which makes me think it's a year's worth of data, maybe tracking daily habits.

Looking at the averages: people are around 38.6 years old, sleep about 6.97 hours, which is a bit low since the recommended is 7-8. Sleep quality is 5.5 out of 10, so not great. Exercise is 44 minutes a day, which is good, but caffeine intake is 146 mg, which is moderate. Screen time before bed is 91 minutes, which might interfere with sleep. Work hours average 8 hours, but productivity is 5.6, mood is 5.4, and stress is 5.5—these are all mid-range scores, indicating room for improvement.

Trends: Older age correlates with higher productivity (median 6 vs 8). More sleep hours (up to 8.21) and better quality (8) are linked to higher productivity and mood. Exercise and lower screen time also relate to better outcomes. Anomalies include people sleeping less than 5 hours but still having high productivity, which might be due to efficiency, and some with high screen time but good sleep quality, maybe due to screen habits.

Business insights: Promote better sleep hygiene by reducing screen time before bed. Encourage exercise and moderate caffeine. Address work-life balance since high work hours might increase stress. Target older employees for productivity strategies. Maybe there's a need for stress management programs. The summary should condense all this into 10 lines, highlighting key points.

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Llama-3.3-70B

=====

Time Taken : 0.97 seconds

The data tracks daily habits of 5000 people, including sleep, exercise, and work.
Average sleep is 7 hours, with 6-9 hours being the norm.
Most people exercise for 44 minutes/day and consume 146mg of caffeine.
Screen time before bed averages 91 minutes, which may affect sleep quality.
Trends show that more sleep and exercise correlate with higher productivity and mood scores.
Anomalies include people sleeping less than 5 hours or exercising over 90 minutes/day.
Business insight: promoting healthy sleep and exercise habits can boost productivity.
Average work hours are 8 hours/day, with a productivity score of 5.37.
Stress levels are moderate, with an average score of 5.55.
Overall, the data suggests that a balanced lifestyle is key to well-being and productivity.

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summary of csv

GPT-OSS-120B

Time Taken : 3.1 seconds

1. This dataset contains 5,000 daily records for 5,000 unique people (≈ 1 year of data) with demographics, sleep, activity, caffeine, screen-time, work, productivity, mood, and stress variables.
2. Age averages 38.6 years (range 18-59, $\sigma \approx 12$), with a fairly even spread across the adult age-range.
3. Sleep: average start 22:00, end $\approx 5:00$ am, total 6.97 h (25-75 % = 5.69-8.21 h); many participants fall short of the recommended 7-9 h.
4. Exercise is recorded as only ≈ 5.5 minutes/day (median 5 min, 75 % = 8 min) – unusually low and likely a unit-recording issue.
5. Caffeine intake averages 44 mg ($\approx \frac{1}{2}$ cup coffee) but spikes to 299 mg; high variability ($\sigma \approx 86$ mg) suggests some heavy users.
6. Screen time before bed averages 146 min (≈ 2.4 h) with a max of 179 min, which can erode sleep quality and total sleep hours.
7. Work hours average 8 h/day (range 4-12 h); longer work correlates with lower productivity scores (avg 5.64, max 11.99) and higher stress.
8. Productivity, mood, and stress scores sit around the mid-scale (≈ 5 -6/10); stress tends to rise when sleep < 7 h, caffeine > 150 mg, or screen time > 60 min.
9. Anomalies: the “Exercise (mins/day)” column tops out at 10 min, which is implausibly low—likely a data-entry or unit error; also a few sleep-start times near 24:00 (23.98) hint at

models time

